

3.2 Ionic bonding homework — answers

Section A — Multiple choice

A1 = B

A2 = C

A3 = B

A4 = C

A5 = D

A6 = B

A7 = C

Section B — Mark scheme clips

B1(a) [2]

Question	Answer	Marks
1(a)	M1 ionic = oppositely charged ions / cations AND anions	1
	M2 covalent = shared pair(s) of electrons AND two nuclei	1

B2(a)(i) [1]

2(a)(i)	Li ⁺ AND S ²⁻	1
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B2(a)(ii) [2]

2(a)(ii)	M1 giant	1
	M2 (many) strong force(s) of attraction between oppositely charged ions OR (many) strong ionic bond(s)	1

B3(a)(i) [1]

Question	Answer	Marks
1(a)(i)	sodium – metallic AND sodium oxide – ionic	1

B3(a)(iii) [1]

1(a)(iii)	(sodium and oxide) ions move (through the liquid)	1
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B4(a)(i) [1]

Question	Answer	Marks
3(a)(i)	NaCl AND MgCl ₂	1

B5(c)(ii) [2]

2(c)(ii)	- false - Al₂O₃ melting point is lower than MgO (but) charge on Al ion is greater than that on Mg	2
	Any 2 points = 1 mark , all 3 points = 2 marks	